# Architecture Decision Records

*Software Design II - Universidad del Norte | Project 5 / RFP DH-05 - Internship and Practicum Management System*

*Evaluation 2 - Architecture: justifying the solution structure | Team: Sebastián Mercado - Cher Jiménez - Juan Arbeláez*

An ADR preserves the reasoning behind a consequential choice: what we knew, what we decided, what we rejected, what we accepted as a consequence, how the decision can be challenged and when it must be revisited. ADRs are historical evidence: a later decision never rewrites an earlier one, it supersedes it with a new record. The records follow template A.1 of the Evaluation 2 guide; ADR-000 also follows the template supplied with the starter kit. ADR-010 was added after reviewing the first slice.

| ID | Title | Status | Drivers |
|---|---|---|---|
| ADR-000 | Adopt the Horizon common starter kit as technical baseline | Selected | C-01, C-03, C-06, AD-05 |
| ADR-001 | Use a modular monolith as architectural style | Selected | AD-01, AD-02, AD-03, AD-05, C-01, C-03 |
| ADR-002 | One relational database with module-owned tables and transactional commands | Selected | AD-01, AD-03 |
| ADR-003 | Explicit placement lifecycle with gated transitions | Selected | AD-01, AD-03 |
| ADR-004 | Versioned rule configuration evaluated through Strategy | Selected | AD-04, QR-05 |
| ADR-005 | Centralized relationship-based access with audience projections | Selected | AD-02, QR-01, QR-02 |
| ADR-006 | Append-only audit history and versioned records | Selected | AD-03, QR-03 |
| ADR-007 | Ports and adapters for identity, student information and employer directory | Selected | AD-05, QR-04, ASM-002, ASM-003 |
| ADR-008 | In-process domain events with a transactional outbox for side effects | Selected | AD-05, QR-04, AD-03 |
| ADR-009 | Do not adopt microservices, service mesh, serverless, low-code or RAG in this release | Selected | AD-01 to AD-05, C-01, C-05 |
| ADR-010 | Concurrent commands: conditional update on status and a correlation id per command | Selected | AD-01, AD-03 |

# ADR-000 - Adopt the Horizon common starter kit as technical baseline

| | |
|---|---|
| **Status** | Selected |
| **Date / owner** | 2026-09-24, revised 2026-10-11 / Sebastián Mercado (architecture owner) |
| **Drivers** | C-01, C-03, C-06, AD-05 |
| **Context** | The course supplies horizon-common-starter: a FastAPI application with environment configuration, a generic SQLAlchemy table (imported_records), CSV and JSON importers with validation reports, a mock identity service, a mock Student Information Service and unit, integration and importer tests. Its README states that it does not define the domain model, workflow, authorization, schema or architecture. We also received the DH-05 discovery data pack with two service contracts (Student Information, Employer Directory) and their failure responses. |
| **Decision** | Keep: FastAPI, the configuration mechanism, the importers and validation scripts, the interfaces / application / domain / infrastructure split, the mock identity header and the test setup (pytest, ruff).<br><br>Modify: the identity mock reads the DH-05 file mock_identities.json; the Student Information mock follows the DH-05 contract route and returns the three documented failures; PROJECT_CODE becomes DH-05.<br><br>Add: an Employer Directory mock, the business modules of ADR-001, domain tables per module, the audit and outbox tables.<br><br>Replace: the generic imported_records table is kept only as a staging area for raw evidence; the domain lives in module-owned tables built by a documented transformation. |
| **Alternatives considered** | (1) Start from an empty repository with another stack. (2) Use the starter unchanged and model the domain inside imported_records. |
| **Rationale** | The starter already runs, is tested and matches the size of the team (C-01). Python and FastAPI are known to the team and simple to maintain (C-03). Starting from zero spends the first sprint on setup. Using imported_records as the domain would copy the export structure, which the data pack and ASM-010 forbid. |
| **Consequences** | We inherit the starter conventions and must keep its tests passing. We own the transformation from raw exports to domain tables and must document it (data/generated/transformation_log.md). Python 3.12 is required on every team computer. |
| **Verification** | Starter tests still pass (tests/importers, test_health_endpoint.py, test_sample_endpoint.py); the DH-05 files are used by the adapters. |
| **Reconsideration / supersession** | If the chosen web technology for Evaluation 3 cannot consume a FastAPI backend, or if a requirement needs a capability the stack cannot offer. |
| **Trace links** | C-01, C-03, C-06; EU-04; repository README |

# ADR-001 - Use a modular monolith as architectural style

| | |
|---|---|
| **Status** | Selected |
| **Date / owner** | 2026-09-24, revised 2026-10-11 / Sebastián Mercado (architecture owner) |
| **Drivers** | AD-01, AD-02, AD-03, AD-05, C-01, C-03 |
| **Context** | Eleven modules (M-01 to M-11) share one placement lifecycle. The most important driver (AD-01) needs the readiness conditions, the approvals and the placement status to be consistent at the moment of activation. Expected load is small (F-22). The team has three people and one semester. |
| **Decision** | Build one deployable backend application organized in business modules with explicit public interfaces. Modules own their data and may only call other modules through those interfaces or react to their domain events. The same artifact runs as API process and as worker process (outbox and reminders). |
| **Alternatives considered** | Layered monolith: gives the local transaction but does not stop every layer from talking to every other, so module boundaries decay. Microservices: would turn the activation gate into a distributed transaction or saga for 600-900 placements a year and a team of three. Low-code platform: does not fit relationship-based access, versioned rules or the simulated service contracts. Full comparison in the Alternatives sheet. |
| **Rationale** | Local transactions protect the gates and the audit history. Module boundaries prevent the 'everything talks to everything' decay and let each team member own modules. One pipeline and one database fit C-01 and C-03. |
| **Consequences** | Everything scales together, which is acceptable at this load. Module boundaries are a discipline: we enforce them with the dependency rules of the Module Design sheet and with an automated architecture test that reads the imports of the code and fails the test suite when a module reaches into another one outside the allowed table. |
| **Verification** | tests/architecture/test_module_boundaries.py fails when a module imports another one outside the allowed table; one runnable artifact (API + worker). |
| **Reconsideration / supersession** | Independent teams take ownership of modules; notification or reporting volume degrades the core; a module needs a different release cadence; a failure in one module must be isolated at runtime. |
| **Trace links** | AD-01, AD-02, AD-03, AD-05; V-02, V-03; QS-08; TP-08 |

# ADR-002 - One relational database with module-owned tables and transactional commands

| | |
|---|---|
| **Status** | Selected |
| **Date / owner** | 2026-09-24, revised 2026-10-11 / Sebastián Mercado (architecture owner) |
| **Drivers** | AD-01, AD-03 |
| **Context** | The activation command must read readiness, approvals and supervisors and change the status atomically, and the audit entry must never be lost when a change happens. |
| **Decision** | Use one relational database (SQLite in local development, PostgreSQL in the shared environment through docker-compose). Each table belongs to exactly one module; other modules never query it directly. Each command runs in one transaction that includes the state change, the audit entry and the outbox message. |
| **Alternatives considered** | One database per module: rejected because the gate needs state, audit and outbox in one local transaction. A document database: rejected because the data is relational (placements, conditions, supervisors) and integrity rules are easier to keep with transactions and keys. |
| **Rationale** | Strong consistency with no distributed transaction. SQLite keeps the starter runnable; PostgreSQL is already prepared in the starter docker-compose. |
| **Consequences** | Table ownership must be respected by convention and review. Reports read through read-only queries owned by the Reporting module. |
| **Verification** | test_failure_in_the_middle_rolls_everything_back; test_two_concurrent_activations_only_one_commits. PostgreSQL run: NOT EXECUTED (TP-14). |
| **Reconsideration / supersession** | If a module is extracted as a service (ADR-001 triggers). |
| **Trace links** | AD-01, AD-03; QS-04, QS-09; INV-04, INV-05; V-02 |

# ADR-003 - Explicit placement lifecycle with gated transitions

| | |
|---|---|
| **Status** | Selected |
| **Date / owner** | 2026-09-24, revised 2026-10-11 / Sebastián Mercado (architecture owner) |
| **Drivers** | AD-01, AD-03 |
| **Context** | Today activation and start are merged (OB-07), completion is judged only by hours (OB-11), and the exports mix states with consequences ('Blocked', 'Activation pending'). The lifecycle has few states but two critical guards. |
| **Decision** | The Placement aggregate has an explicit transition table: PENDING_ACTIVATION -> ACTIVE -> COMPLETION_PENDING -> COMPLETED -> CLOSED, plus ACTIVE <-> ON_HOLD, CLOSED with result INTERRUPTED from ACTIVE, ON_HOLD or COMPLETION_PENDING, and CANCELLED from PENDING_ACTIVATION. Two transitions have guards: activation requires the readiness gate; completion requires the closure gate (accepted hours, final evaluations without hold, academic signoff). The real start date is an attribute recorded after activation, not a state. 'Blocked' is derived from the gate, not stored. Activation is written with a conditional update on the status that was read (ADR-010), and the other commands are designed to reuse it. The 'result INTERRUPTED' attribute and the closure gate are designed; the code today implements the transition table and the activation gate. |
| **Alternatives considered** | State pattern (one class per state); status as free text as in the exports; a BPM engine. |
| **Rationale** | A transition table is the simplest structure that makes illegal transitions impossible and testable. The State pattern adds one class per state while the behavior per state is still small. |
| **Consequences** | Any new state requires updating the table and its tests. If behavior per state grows (for example per placement type), we revisit the State pattern. |
| **Verification** | tests/unit/test_placement_lifecycle.py (6); activation tests (8). Closure gate: DESIGNED only. |
| **Reconsideration / supersession** | When states start to carry different behavior, or when CL-011 (which changes during a placement need a new approval, still open from Evaluation 1) defines changes that restart the academic review. |
| **Trace links** | AD-01; QS-08; BR-01, BR-03, BR-04, BR-06; V-04; lifecycle figure |

# ADR-004 - Versioned rule configuration evaluated through Strategy

| | |
|---|---|
| **Status** | Selected |
| **Date / owner** | 2026-09-24, revised 2026-10-11 / Sebastián Mercado (architecture owner) |
| **Drivers** | AD-04, QR-05 |
| **Context** | Eligibility combines opportunity criteria (eligible programs, minimum credits, minimum average) with program rules; the observed reasons for ineligibility are enrollment status (ELIG-5005) and academic standing (ELIG-5006). Readiness conditions vary by placement type and program (CL-020, EU-02). Rules change between periods and the office has few technical people. |
| **Decision** | Rules are data: a versioned configuration file (config/rules/readiness.json, version readiness-v1) lists which rules apply per placement type, optionally per program (lookup 'TYPE@PROGRAM', then 'TYPE', then 'default'), with their parameters. Each rule type is a small evaluator implementing one interface (Strategy); a factory builds the evaluators from the configuration. Implemented readiness rule types: REQUIREMENT_COMPLETE, AGREEMENT_VALID_IN_DIRECTORY and SUPERVISORS_ASSIGNED. Eligibility and completion rule types (PROGRAM_IN_LIST, MIN_CREDITS, MIN_AVERAGE, ENROLLMENT_ACTIVE, STANDING_ALLOWED, ACCEPTED_HOURS_AT_LEAST, EVALUATION_SUBMITTED) are DESIGNED. Every gate decision records the rule version in its audit reason ('readiness gate readiness-v1'); a refusal also stores each blocking rule result in the audit new_value. A dedicated rule_version column on decisions is DESIGNED. |
| **Alternatives considered** | Hard-coded conditionals; an external rule engine (Drools-like); a full BPM tool. |
| **Rationale** | Adding or changing a condition is a configuration change (QS-07). New rule types are new classes that do not modify the callers (open/closed). Storing the version answers auditability questions (AD-03). |
| **Consequences** | Somebody must own rule governance (who may publish a new version): recorded as open question for CL-004 and CL-020. The first release edits configuration files reviewed by the team; an admin screen is a later backlog item. |
| **Verification** | test_rule_change_is_configuration_only; test_qs04_audit_reason_records_the_rule_version_used. Eligibility rules: DESIGNED. |
| **Reconsideration / supersession** | If rules need logic that cannot be expressed as a combination of rule types. |
| **Trace links** | AD-04; QS-07; BR-08; EU-01, EU-02; V-04 |

# ADR-005 - Centralized relationship-based access with audience projections

| | |
|---|---|
| **Status** | Selected |
| **Date / owner** | 2026-09-24, revised 2026-10-11 / Sebastián Mercado (architecture owner) |
| **Drivers** | AD-02, QR-01, QR-02 |
| **Context** | Roles are not enough: a workplace supervisor may see only their own placements, a host organization only its opportunities and candidates, and one evaluation mixes student-visible feedback with restricted comments. CL-018 (who may read restricted content) is still open. |
| **Decision** | An Access module is the only place where access is decided. Application services call check(actor, action, placement relations) before any read or command. The decision uses the role of the user and their relationship with the resource (student of, contact of the organization, workplace supervisor of, faculty supervisor of, office staff with role EXPERIENTIAL_LEARNING_COORDINATOR, PROGRAM_DIRECTOR or AGREEMENTS_REPRESENTATIVE; 'coordinator of the program' is DESIGNED). Responses are built through audience projections (student, host organization, workplace supervisor, faculty, office) that include only the allowed fields. Default is deny; until CL-018 is closed, restricted fields stay limited to the office, the faculty supervisor of the placement and the author of the content (a workplace supervisor reads their own comments). |
| **Alternatives considered** | Role checks in each endpoint; protection proxy around repositories; database row-level security. |
| **Rationale** | One decision point is easier to test exhaustively (QS-02, QS-03) and to change when CL-018 is answered. Projections make field-level privacy explicit instead of relying on the UI to hide fields. |
| **Consequences** | Every new endpoint must declare its action and projection; tests cover the access matrix. Relationship data (assignments, contacts) becomes critical and is audited. |
| **Verification** | test_unrelated_user_gets_not_found; test_host_organization_sees_readiness_without_restricted_evidence; test_qs02_*. Evaluation/incident projections: DESIGNED. |
| **Reconsideration / supersession** | When CL-018 is closed, or if the number of audiences grows beyond what projections can express simply. |
| **Trace links** | AD-02; QS-01, QS-02, QS-03; BR-09, BR-10; T-02, T-03, T-08 |

# ADR-006 - Append-only audit history and versioned records

| | |
|---|---|
| **Status** | Selected |
| **Date / owner** | 2026-09-24, revised 2026-10-11 / Sebastián Mercado (architecture owner) |
| **Drivers** | AD-03, QR-03 |
| **Context** | Hours are disputed (HOUR-002), work plans are revised (WP-001 is v2) and decisions must be reconstructed (OB-12). Editing records in place loses the previous value. |
| **Decision** | An Audit History module stores one entry per state-changing command: entry id, time, actor, action, resource, previous value, new value, reason and correlation id. Entries are written by the command in the same transaction and there is no update or delete operation. Work plans, eligibility decisions, hour reviews and readiness evidence are new records or versions linked to the previous one. |
| **Alternatives considered** | Database triggers; logging to files; full event sourcing. |
| **Rationale** | Explicit entries written by the application carry the business reason and the actor, which triggers do not know. Event sourcing would solve the same problem at a much higher complexity for this team. |
| **Consequences** | Storage grows with every change, which is small at this volume. Reports of history are queries on one table. The application exposes no update or delete, but a database administrator could still alter rows: this is recorded as risk AR-05. One correlation id links the entries of a command (ADR-010). |
| **Verification** | test_audit_history_can_only_be_appended_and_read; audit assertions in activation tests; test_one_correlation_id_links_audit_entry_and_notifications. |
| **Reconsideration / supersession** | If an external audit store or legal retention policy is required. |
| **Trace links** | AD-03; QS-04; BR-07; INV-06, INV-07; T-05, T-07 |

# ADR-007 - Ports and adapters for identity, student information and employer directory

| | |
|---|---|
| **Status** | Selected |
| **Date / owner** | 2026-09-24, revised 2026-10-11 / Sebastián Mercado (architecture owner) |
| **Drivers** | AD-05, QR-04, ASM-002, ASM-003 |
| **Context** | The Student Information Service is authoritative for program, credits, average, standing and enrollment; the Employer Directory is authoritative for organization identity and agreement summary. Both are simulated and fail with 404, 409 and 503. The starter identity mock uses a different file name (EU-04). |
| **Decision** | The core defines ports (StudentInformationPort, EmployerDirectoryPort, IdentityPort) in the shared kernel (horizon/shared/ports.py), so that any module can depend on a port without depending on M-11. Adapters, which live only in M-11, implement them for the mocks and translate their payloads into internal types, recording the source timestamp. Failure behavior is decided per use case: eligibility check -> keep the application as ELIGIBILITY_PENDING and retry (fail soft); 409 stale profile -> manual review; activation agreement check -> refuse the activation (fail closed); unknown organization or student (404) -> business error shown to the user. Values returned by a port (StudentProfile, AgreementSummary) are transient snapshots used for one decision; they are not stored as master data, and DH-05 never becomes the authoritative source. |
| **Alternatives considered** | Call the mock endpoints directly from services; copy student and employer data into our tables as master data. |
| **Rationale** | The contracts say the internship system must not become the authoritative academic-record or agreement repository. Ports keep the core testable with fake adapters and let the university replace the mocks later. |
| **Consequences** | Each adapter needs contract tests and failure tests. Values read may be stale; the port values carry read_at, and recording it with each decision is DESIGNED. |
| **Verification** | tests/unit/test_reference_and_adapters.py; test_directory_down_means_activation_refused; architecture test: mocks imported only by M-11 and composition points. |
| **Reconsideration / supersession** | When real institutional services are integrated, or if a service adds events/webhooks. |
| **Trace links** | AD-05; QS-05, QS-08; BR-02; FM-02..FM-05; TB-2 |

# ADR-008 - In-process domain events with a transactional outbox for side effects

| | |
|---|---|
| **Status** | Selected |
| **Date / owner** | 2026-09-24, revised 2026-10-11 / Sebastián Mercado (architecture owner) |
| **Drivers** | AD-05, QR-04, AD-03 |
| **Context** | One decision triggers several reactions: an activation must notify the student and both supervisors, update reminders and feed reports; an academic approval completes a readiness condition. A failed notification must never cancel the decision (ASM-009). |
| **Decision** | Modules publish domain events (for example AcademicApprovalRecorded, WorkPlanApproved, PlacementActivated, HoursReviewed, PlacementClosed) through an in-process dispatcher (Observer). Handlers that change business data run in the same transaction. Handlers with external side effects (notifications; reminders are DESIGNED) only write to an outbox table; the worker process delivers them and retries failures. |
| **Alternatives considered** | Call every reaction directly from the service; external message broker; serverless functions. |
| **Rationale** | The service only announces what happened (open/closed), and side effects are isolated from the transaction. The outbox gives reliability without new infrastructure. |
| **Consequences** | Delivery of notifications is eventual. The team must keep the event catalog (Interfaces sheet) up to date. |
| **Verification** | tests/integration/test_outbox.py; test_qs06_worker_crash_*; test_qs06_outbox_payload_*. |
| **Reconsideration / supersession** | If a module is extracted as a service, the same events move to a broker (ADR-001 triggers). |
| **Trace links** | AD-05; QS-06; BR-11; FM-11, FM-12; T-04 |

# ADR-009 - Do not adopt microservices, service mesh, serverless, low-code or RAG in this release

| | |
|---|---|
| **Status** | Selected |
| **Date / owner** | 2026-09-24, revised 2026-10-11 / Sebastián Mercado (architecture owner) |
| **Drivers** | AD-01 to AD-05, C-01, C-05 |
| **Context** | The course presented several technologies. Each was run through the decision filter: problem, driver, simpler alternative, benefit, cost, evidence. |
| **Decision** | Not adopted now: microservices, service mesh, external broker, serverless functions, low-code platform, RAG assistant. Adopted instead: modular monolith, outbox, a worker process (run on demand today; scheduled reminders designed), custom UI over the API, rule text shown in the UI. |
| **Alternatives considered** | Adopt any of them now. |
| **Rationale** | None solves a problem that our drivers define, and each adds cost the team cannot absorb (C-01). AI may not decide eligibility or select candidates (RFP section 10, C-05). |
| **Consequences** | We record the triggers under which each option becomes justified (Decision Filter sheet), so the decision can evolve with evidence. |
| **Verification** | Not a testable decision; reviewed against the decision filter. Revisit triggers recorded in ADR-001. |
| **Reconsideration / supersession** | Triggers listed in ADR-001 and in the Decision Filter sheet. |
| **Trace links** | AD-01..AD-05; C-01, C-05; Decision Filter |

# ADR-010 - Concurrent commands: conditional update on status and a correlation id per command

| | |
|---|---|
| **Status** | Selected |
| **Date / owner** | 2026-10-05, revised 2026-10-11 / Sebastián Mercado (architecture owner) |
| **Drivers** | AD-01, AD-03 |
| **Context** | Review of the first slice found a hole: two coordinators can read the same PENDING_ACTIVATION placement, both pass the gate and both write ACTIVE, leaving two success audit entries and two sets of notifications. ADR-002 gives a transaction, but a transaction alone does not stop a lost update. Also, the audit entry, the event and the outbox messages of one command had no common identifier, so QS-04 (reconstruct a decision) was harder than it should be. |
| **Decision** | The repository saves a placement with an UPDATE … WHERE status = <status that was read>. If no row changes, it raises StaleState and the command rolls back; the API answers 409 with code STATE_CHANGED. Every command generates one correlation id, written to its audit entries, carried by the domain event and copied to the outbox messages and to the API response. Error responses carry a machine-readable code (ACTIVATION_BLOCKED, STATE_CHANGED, ILLEGAL_TRANSITION). |
| **Alternatives considered** | Pessimistic row lock (SELECT … FOR UPDATE: not supported by SQLite and needs more care with the gate's external call); version column (optimistic locking with a counter: heavier schema change for the same effect here); a unique constraint on 'activation' records; do nothing and rely on the low load. |
| **Rationale** | Conditional update on the status that the command already depends on needs no new column, is supported by both SQLite and PostgreSQL (tested on SQLite only, TP-14), and makes the lost update impossible instead of unlikely. The rule 'activate only from PENDING_ACTIVATION' already existed; this makes the database enforce it at write time. |
| **Consequences** | Callers must handle 409 STATE_CHANGED. The gate's external call happens before the write, so the loser may have called the directory for nothing: acceptable. The same pattern must be reused by every status-changing command (hold, resume, complete, close). We tested it with a deterministic interleaving on SQLite, not with parallel connections on PostgreSQL (TP-14). |
| **Verification** | test_two_concurrent_activations_only_one_commits; test_one_correlation_id_links_audit_entry_and_notifications; test_refusal_and_denial_carry_machine_readable_codes. Parallel PostgreSQL run: NOT EXECUTED. |
| **Reconsideration / supersession** | If a command needs to change more than the status atomically (then use a version column), or if contention becomes frequent. |
| **Trace links** | AD-01, AD-03; QS-09; BR-12; INV-04, INV-07; FM-08; T-09; SEQ-04 |